

% *Expresii Lambda in Prolog*

% generare permutari si submultimi

**sterge1(X,[X|T], T).
sterge1(X, [Y|T], [Y|TR]) :- sterge1(X,T,TR).**

**permutari([],[]).
permutari(L,[X|P]) :- sterge1(X,L,L1),permutari(L1,P).**

**submultimi([],[]).
submultimi([H|T],[H|TR]) :- submultimi(T,TR).
submultimi([H|T],L) :- submultimi(T,L).**

% sortarea prin interclasare

**intersort([],[]).
intersort([H],[H]).
intersort([H1,H2|TR],LS) :-
 split([H1,H2|TR],L1,L2),
 intersort(L1,LS1),
 intersort(L2,LS2),
 interclasare(LS1,LS2,LS).
split([],[],[]).
split([H],[H],[]).
split([H1,H2|TR],[H1|TR1],[H2|TR2]):-
 split(TR,TR1,TR2).
interclasare(L,[],L).
interclasare([],L,L).
interclasare([H1|TR1],[H2|TR2],[H1|TR]) :-
 H1 < H2, interclasare(TR1,[H2|TR2],TR).
interclasare([H1|TR1],[H2|TR2],[H2|TR]) :-
 H1 > H2, interclasare([H1|TR1],TR2,TR).**

**% maplist(_,Goal, List1, List2) :-
% maplist(_,List1, List2, Goal).**

**% maplist(_,[], [], _).
% maplist(_,[Elem1|Tail1], [Elem2|Tail2], Goal) :-
% call(Goal, Elem1, Elem2),
% maplist(_,Tail1, Tail2, Goal).**

**% maplist(_, [], []).
% maplist(Pred, [Old|Olds], [New|News]) :-
% apply(Pred, [Old,New]),**

% maplist(Pred, Olds, News).

patrat(A,R):- R is A*A.

suma1(X,Y,Z):- Z is X + Y.

sumaL([],0).

sumaL([H|T],S):- sumaL(T,S1), S is S1+H.

cmmdc(X,Y,Y):- mod(X,Y)=:0,!.

cmmdc(X,Y,D):- R is mod(X,Y), cmmdc(Y,R,D).

cmmdcL([H],H).

cmmdcL([H|T],D):- cmmdcL(T,D1), cmmdc(D1,H,D).

% maplist(patrat,[2,3],L).

% maplist(suma1,[1,3],[4,6],R).

% forall(member(X,[1,2,3]),

% (Y is X+1, writeln(Y))).

add1([],[], K).

add1([H|T],[R|TR], K):- R is H+K, add1(T,TR,K).

% maplist(sumaL,[[2,3],[11,2],[4,7]],L).

% maplist(cmmdc,[20,70,7,3],[15,80,7,6],L).

% maplist(cmmdcL,[[20,30,50],[21,7],[40,25]],L).

cmmdc2el:- call(maplist(cmmdc,[20,30,50],[60,40,25],L)),write(L).

cmmdcLL:-

call(maplist(cmmdcL,[[20,30,50],[21,7],[40,25]],L)),write(L).

intersectie([],L,[]).

intersectie([H|T],L,[H|TR]):- member(H,L), intersectie(T,L,TR),!.

intersectie([H|T],L,R):- intersectie(T,L,R).

intersectieL:-

call(maplist(intersectie,[[20,40,50],[21,7,3,40],[3,4,5,6]],[[40,25],[40,54,3],[5,6,7,8]],L)),write(L).

intersortLL:- call(maplist(intersort,[[20,-3,5,-7],[2,1,7,-3],[4,0,2,-5]],L)),write(L).

submultimeLL:- call(maplist(submultimi,[[20,-3],[2,1,7],[4,0,2]],L)),write(L).

permutariLL:- call(maplist(permutari,[[20,-3],[2,1,7],[4,2]],L)),write(L).